

Object Counting By Tracking For Complex Video Sequences

Jera Makar, Hani Hagrass
The Artificial Intelligence Group
School of Computer Science and Electronic Engineering,
University of Essex
Email: Jeramakar@gmail.com, hani@essex.ac.uk

Hugo Leon-Garza, Anasol Pena-Rios
Business Modelling and Operational Transformation Practice
British Telecom
Ipswich, UK
Email: hugo.2.leongarza@bt.com, anasol.penarrios@bt.com

Abstract—Object counting within videos plays a crucial role in numerous applications, including crowd monitoring, traffic analysis, surveillance, and resource management. The ability to reliably count and track objects is critical for ensuring safety in public spaces, managing traffic flows, and analyzing behavior patterns in various domains. However, the task of counting objects in video frames poses significant challenges, including overlapping objects, occlusions, and repetitive counting across multiple frames. Previous methods such as counting-by-detection, regression, and segmentation have demonstrated limited effectiveness in these complex scenarios. Specifically, it was shown that counting-by-detection struggles with overlapping objects, while regression-based approaches discard crucial spatial information, and segmentation techniques depend heavily on image quality and struggle with occlusions.

This paper addresses the abovementioned challenges and presents a new system encompassing advanced object detection, re-identification, and multi-object tracking algorithms, including the YOLOv10 model and ByteTrack. The proposed system performance was further enhanced by adding a custom Non-Maximum Suppression (NMS) function and a minimum frame count threshold (MIN_FRAMES) to reduce false positives. We conducted several experiments which showed improved accuracy and reliability of the tracking system, particularly in challenging environments with occlusions and multiple appearances, leading to 91.43% average improvement in RMSE when compared to the state-of-the-art Ultralytics object counting method and a 60.53% average improvement in RMSE compared to not adding the NMS and MIN FRAMES to our system.

Keywords: Video analysis, Object counting, Multi-object tracking

I. INTRODUCTION

The increasing volume of video data in fields like entertainment, security, industrial monitoring, and scientific research has driven the demand for automated tools to analyze and extract insights. Among the critical tasks in video analysis is object counting—estimating the number of objects in video frames. This task has diverse applications, from counting cells in biomedical imagery [1] to monitoring crowds [2], analyzing traffic (shown in Fig. 1) and even estimating the number of trees in aerial photography [3]. Several approaches have been developed to tackle this challenge, each with its own strengths and limitations. The **counting-by-detection** approach leverages object detection algorithms to locate individual objects but struggles with overlapping or occluded objects. Generative methods model object relationships to address these issues but are computationally demanding. Meanwhile, **counting by regression** [4] maps global image features to object counts, bypassing detection but losing spatial information and requiring extensive labeled data. Finally, **counting by segmentation** [5] clusters similar objects before estimating counts, balancing

accuracy and efficiency, though it performs poorly in low-quality images or noisy environments. This research focuses on the counting-by-detection approach, aiming to build a robust system for challenging scenarios like occlusions (shown in Fig. 2), where objects are blocked or appear multiple times in different frames (shown in Fig. 3.). A sophisticated Multi-Object Tracking (MOT) system is employed to ensure object identities are maintained across frames. According to [7], an ideal MOT system must track the correct number of objects, estimate their positions accurately, and generate one trajectory per object, even under occlusions.

The proposed system integrates advanced object detection (YOLOv10), multi-object tracking (ByteTrack), and re-identification techniques. Enhancements, such as a custom Non-Maximum Suppression (NMS) function and a minimum frame count threshold (MIN_FRAMES), further reduce false positives. The paper outlines the problem (Section II), the proposed counting-by-tracking technique (Section III), experimental results (Section IV), and conclusions with future directions (Section V). Due to an agreement of disclosure between the authors and BT, the code used in this study cannot be made publicly available.

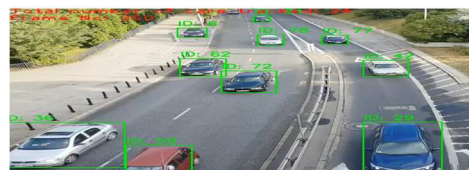


Fig. 1. A screenshot of our system counting on real world video data [7] where the system counts 24 cars and the ground truth is 24.

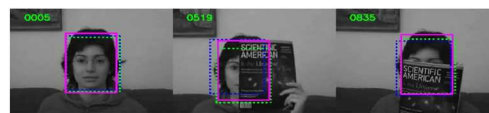


Fig. 2. Example of occlusion [8]



Fig. 3. Example of multiple instance appearances of the same object in different frames [8].

II. OVERVIEW OF THE OBJECT COUNTING PROBLEM IN VIDEOS

A. Multi-Object Tracking Task

Multi-Object Tracking (MOT) is essential for preserving object identities across video frames, ensuring accurate

counting. The process begins with object detection, where models like YOLO [9], R-CNN [10], and Faster R-CNN [11] locate and classify objects using bounding boxes. Preprocessing techniques such as resizing and normalization enhance detection performance. Features like appearance traits as color and shape and motion attributes like speed and direction are then extracted to maintain consistent tracking of objects. Data association methods, such as the Hungarian algorithm [12], pair detections with tracks by minimizing positional differences, while Kalman filters [13], predict future states to handle temporary occlusions. Finally, post-processing refines object trajectories by eliminating irregular tracks, interpolating missing data, and correcting identity switches, ensuring smooth and accurate results.

B. Multi-Object Tracking Algorithms

Multi-Object Tracking (MOT) algorithms typically follow one of three approaches. Tracking-by-Detection separates detection and tracking steps, first identifying objects and then associating them across frames. Joint-Detection-and-Tracking combines these processes into a single system, improving efficiency. Detection-Free methods, on the other hand, bypass object detection entirely, focusing exclusively on tracking. Each approach offers a different balance between accuracy, speed, and robustness, catering to various video analysis needs.

1) Tracking-by-Detection Algorithms

SORT[14] is a real-time tracking method that uses Intersection-over-Union (IoU) distance for object association, focusing on speed. It employs a Kalman filter for motion prediction and the Hungarian algorithm for track assignment, relying on Faster R-CNN for detection. DeepSORT[15] improves upon SORT by integrating appearance features with motion data, using Convolutional Neural Networks (CNNs) for person re-identification, which enhances robustness in occluded scenarios. StrongSORT[16] further builds on DeepSORT by incorporating advanced embeddings and association modules, utilizing a Bag of Tricks (BoT) [17] feature extractor based on a ResNeSt50 backbone pre-trained on the DukeMTMC dataset [18]. It also employs Exponential Moving Average (EMA) for smoother updates and global linear assignment for better performance. OC-SORT [19] introduces observation-centric methods (ORU and OCM) to address occlusions and nonlinear movements. Deep-OC-SORT[20] combines motion and appearance cues, adding Camera Motion Compensation (CMC), Dynamic Appearance (DA), and Adaptive Weighting (AW) for robust tracking. Hybrid-SORT [21] integrates spatial and appearance cues with weaker cues like velocity, while also incorporating Tracklet Confidence Modeling and Height-Modulated IoU for improved tracking accuracy. ByteTrack [22] enhances tracking accuracy by including low-confidence detections, making it effective for tracking occluded or blurred objects. Finally, BoT-SORT [23] refines bounding box calculations and integrates IoU-based localization with Re-ID, enabling more accurate tracking in dynamic camera scenarios.

2) Joint-Detection-and-Tracking Algorithms

YOLOTracker [24] combines object detection and tracking into a unified framework using YOLOv5. It integrates features through PANet to handle varying object sizes and employs a Kalman filter and the Hungarian algorithm for accurate and consistent tracking in complex environments.

C. Re-Identification Task

Re-Identification (Re-ID) plays a vital role in tracking objects across non-overlapping camera views, overcoming challenges like varying angles, lighting, and occlusions. Deep learning has transformed Re-ID, replacing manual feature crafting with advanced models. Identification models classify identities directly, while verification models, such as Siamese networks [25], evaluate image similarity. Distance metric-based models rank similarities, and part-based models enhance accuracy by combining local and global features. Video-based models leverage temporal and spatial data, while GAN-based augmentation [26] generates synthetic training data to improve generalization. Despite these advancements, practical challenges like background clutter and domain shifts persist, requiring further research to bridge the gap between experimental and real-world applications.

III. THE PROPOSED COUNTING BY TRACKING TECHNIQUE

In this study, we propose a unified pipeline for accurate object counting in video sequences, as illustrated in Fig. 4. The system integrates object detection, re-identification (ReID), and multi-object tracking (MOT) to address challenges like consistent detection, identity maintenance across frames, and preventing double counting. Evaluations across different scenarios using metrics like Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and deviations from ground truth highlight its precision, efficiency, and robustness against occlusion and crowding.

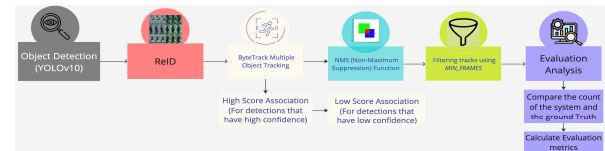


Fig. 4. The pipeline of our Counting By Tracking technique.

A. Object Detection

For object detection, the YOLOv10 [27] architecture was selected for object detection due to its cutting-edge performance, offering a 1.4% improvement in average precision (AP) and up to 38% fewer parameters compared to YOLOv8, making it ideal for real-time video processing. YOLOv10 was trained using pre-trained YOLOv10-S weights with a batch size of 16, an image resolution of 1280×1280 pixels, and the AdamW optimizer. Data augmentation techniques such as mosaic, random erasing, RandAugment, and horizontal flipping were employed to improve robustness. Additional optimizations included mixed precision training, overlap masking, and an IoU threshold of 0.7 to enhance detection accuracy, particularly in crowded scenes.

B. Re-identification (ReID) Models

Object detection alone cannot ensure accurate counting in video sequences due to challenges like objects appearing, disappearing, and reappearing. ReID addresses this by

maintaining identity consistency. The ResNet-50 backbone was chosen for its balance of accuracy and efficiency, extracting high-quality features to distinguish objects. The ReID model, implemented using the BaseReID class from MMTracking framework [29], includes a LinearReIDHead and GlobalAveragePooling neck, producing a 128-dimensional embedding as a unique “fingerprint” for each object. Training combined triplet loss (margin 0.3) to separate object embeddings and cross-entropy loss to enhance classification. Stochastic gradient descent (SGD) was used with a learning rate of 0.1, momentum of 0.9, and a linear warmup for 1000 iterations, followed by multistep decay. Augmentation included random horizontal flipping on the MOT17 ReID dataset. Performance was evaluated with mean Average Precision (mAP) and Cumulative Matching Characteristic (CMC).

C. Multiple Object Tracking Models

Multiple state-of-the-art trackers were implemented, including StrongSORT, OCSORT, DeepOCSORT, ByteTrack, BoT-SORT, and ImprASSoc (also known as YOLOTracker), all sourced from the Boxtmot framework [30]. These trackers were evaluated for their ability to handle occlusions, appearance variations, and camera motion while maintaining tracking accuracy across frames. Among these, ByteTrack demonstrated superior performance, excelling in scenarios with occlusions or motion blur. ByteTrack’s strength lies in its simplicity; it does not rely on ReID weights, significantly reducing computational overhead during training and deployment. ByteTrack employs a two-stage data association process to handle both high- and low-confidence detections, ensuring robust tracking. In the first stage, high-confidence detections are matched with existing tracklets, while the second stage matches low-confidence detections, recovering previously missed objects. This approach enables ByteTrack to maintain accurate tracking even in challenging conditions, making it the preferred choice for the proposed pipeline.

1) High-score association

The first stage of ByteTrack focuses on high-confidence detections, which are more likely to represent accurately identified objects. To determine whether a detection corresponds to an existing track, the system evaluates similarity using either Intersection over Union (IoU) or ReID feature distance, if ReID data is available.

$$\text{Similarity}_1(D_{\text{high}}, T) = \text{IoU}(d_{\text{high}}, t) \quad \text{or} \quad \text{ReID_dist}(d_{\text{high}}, t) \quad (1)$$

Here, D_{high} represents the set of high-confidence detections in the current frame, and T represents the set of established tracks from previous frames. IoU measures the overlap between the bounding boxes of the detected object d_{high} and the tracked object t , providing spatial similarity. ReID feature distance, when available, compares the visual appearance of detected and tracked objects, enabling robust matching under challenging conditions. To optimize matching, the Hungarian algorithm is applied to similarity scores, ensuring each detection is paired with the most appropriate track. This approach reduces errors and ensures the tracking system effectively links high-confidence detections with established tracks.

2) Low-score association

In the second stage, ByteTrack processes low-confidence detections, which are more susceptible to inaccuracies due to factors like occlusion, motion blur, or partial visibility. Since ReID information may be unreliable for these detections, only the IoU metric is used for matching:

$$\text{Similarity}_2(D_{\text{low}}, T_{\text{remain}}) = \text{IoU}(d_{\text{low}}, t_{\text{remain}}) \quad (2)$$

Here, D_{low} is the set of low-confidence detections, and T_{remain} is the set of tracks that were unmatched during the high-score association stage. By computing IoU between remaining tracks and low-confidence detections, the system recovers objects that might have been missed in the first stage due to temporary obstructions or poor detection conditions. Together, the high- and low-score association stages allow ByteTrack to maintain robust tracking performance, even in complex scenarios.

3) Non-Maximum Suppression (NMS) and Enhancements

To further refine detections, ByteTrack incorporates a custom Non-Maximum Suppression (NMS) function, a standard object detection technique used to handle multiple overlapping bounding boxes for the same object. NMS selects the most reliable bounding box based on confidence scores and an IoU threshold:

$$B_{\text{selected}} = \text{NMS}(B, S, \text{IoU threshold}) \quad (3)$$

Here, B is the set of detected bounding boxes, S is the corresponding set of confidence scores, and the IoU threshold determines the overlap level acceptable before suppression. The custom NMS function in ByteTrack is adaptive, ensuring accurate detections are retained while false positives are minimized, thus improving overall tracking precision.

Another enhancement is the introduction of a minimum frame count threshold (MIN_FRAMES). This ensures the system tracks only objects that persist across a minimum number of frames, discarding transient noise or brief appearances.

$$T_{\text{valid}} = \{t \in T \mid \text{count}(t) \geq \text{MIN_FRAMES}\} \quad (4)$$

Here, T_{valid} represents the set of valid tracks, and $\text{count}(t)$ indicates the number of frames an object t has been detected. Objects detected in fewer than the required MIN_FRAMES are excluded, reducing false positive tracks.

Finally, ByteTrack maintains a set of *tracked_ids* for each frame, facilitating real-time visualization of unique objects. This visualization highlights the tracking effectiveness throughout the video sequence, ensuring consistent and reliable object counting in dynamic environments. By combining IoU-based and ReID-based matching, a custom NMS function, and the MIN_FRAMES threshold, ByteTrack achieves superior accuracy and reliability in object tracking and counting. These enhancements allow the system to handle complex challenges like occlusion, motion blur, and crowded scenes, making it highly effective for real-world applications.

IV. EXPERIMENTS AND RESULTS

A diverse set of video datasets, including MOT17 [31] [32] [33] and real-world data from YouTube [7][34], was used to

evaluate the system under challenging conditions such as occlusions and illumination changes. These datasets featured categories like pedestrians in surveillance and vehicles in traffic, with ground-truth counts obtained through manual and pre-annotated annotations. Evaluation intervals ranged from 10 to 50 frames, balancing accuracy and computational efficiency. Error analysis compared predictions with ground-truth using metrics like Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE). TABLE I. summarizes the performance of multi-object trackers on an occluded MOT17 clip, where ByteTrack achieved the lowest average absolute difference (1.05) and RMSE (1.27) with fewer hyperparameters compared to OCSORT and StrongSORT. Enhancements like custom Non-Maximum Suppression (NMS) and the MIN_FRAMES threshold further improved accuracy, achieving a 60.53% RMSE reduction for complex datasets. TABLE II. highlights these enhancements, showing a 91.43% RMSE improvement over the state-of-the-art Ultralytics object counting method [35] across tested sequences.

TABLE I. COMPARISON OF MULTI-OBJECT TRACKERS ON MOT17-04-SDP CLIP DATASET WITH 82 PEOPLE

Tracker	Avg. Abs. Diff.	RMSE	Largest Abs. Diff.	# of Hyperparameters	Needs ReID Weights
ByteTrack	1.05	1.27	2	5	FALSE
OCSORT	2.10	2.56	6	10	FALSE
StrongSORT	2.90	3.42	7	9	TRUE
DeepOCSORT	4.43	4.89	8	17	TRUE
HybridSORT	5.52	7.12	13	11	TRUE
BotSORT	9.14	10.31	20	10	TRUE
ImprAssoc	29.76	36.58	61	15	TRUE

TABLE II. COMPARISON OF BYTETRACK PERFORMANCE WITH AND WITHOUT ENHANCEMENTS

Video	MAE Improvement (%)	RMSE Improvement (%)	Using	Largest Abs. Diff.	Avg. Abs. Diff.	MAE	RMSE
MOT17-0 4-SDP Clip (Ground Truth: 82 people)			Our system with enhancement	2	1.05	1.62	1.27
	92.82%	73.26%	Our system without enhancement	9	4.10	22.57	4.75
	99.87%	96.45%	Ultralytics Object Counting [35]	52	33.9	1286.1	35.8
MOT17-0 9-SDP Clip (Ground Truth: 25 people)			Our system with enhancement	2	0.58	0.73	0.85
	36.52%	20.56%	Our system without enhancement	2	0.85	1.15	1.07
	98.54%	88.03%	Ultralytics Object Counting [35]	11	6.23	50.1	7.1
Shopping, People, Commerce, Mall (Ground Truth: 85 people) [34]			Our system with enhancement	2	1.24	2.06	1.43
	73.28%	48.56%	Our system without enhancement	5	2.41	7.71	2.78
	97.87%	85.41%	Ultralytics Object Counting [35]	13	9.47	96.5	9.8
Road traffic video for object detection (Ground Truth: 51 cars) [7]			Our system with enhancement	4	0.81	1.48	1.21
	87.62%	65.03%	Our system without enhancement	7	2.90	11.95	3.46
	81.93%	57.69%	Ultralytics Object Counting [35]	6	2.48	8.19	2.86

Temporal consistency was evaluated to ensure smooth tracking and avoid duplicate counts, with visual inspections and statistical evaluations identifying issues in occluded or fast-motion scenarios. Error metrics and trends, visualized in Fig. 5., and Fig. 6., provided insights into predicted versus groundtruth counts, guiding system refinements and improving robustness for real-world applications.



Fig. 5. A screenshot of the system counting on MOT17-04-SDP Clip on the 100th Frame. The system's count with enhancements is 36, without enhancements is 48. Ground truth count is 36.

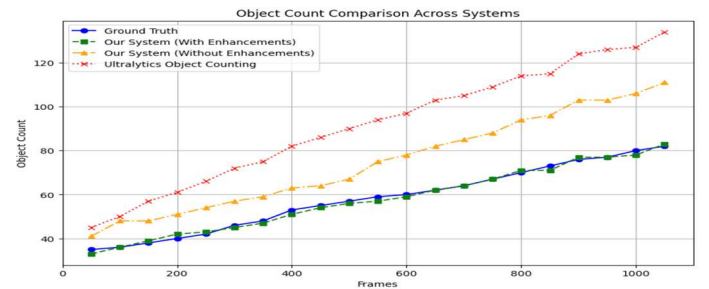


Fig. 6. Line Plots for the MOT17-04-SDP Clip video sequence of our system with enhancements in green, without enhancements in yellow, Ultralytics object counting in red and ground truth in blue.

V. CONCLUSIONS AND FUTURE WORK

This paper presented a robust counting technique capable of handling challenges like occlusions and repetitive counts by integrating object detection, re-identification, and tracking algorithms. The main challenges addressed were managing partially obscured objects and preventing duplicate counts across frames. By combining YOLOv10 for detection, a re-identification model to link objects across frames, and multiple tracking algorithms, the system showed strong performance in video sequences. ByteTrack proved the most effective, excelling in handling occlusions and accurately counting unique objects. Key enhancements included using Non-Maximum Suppression (NMS) to remove overlapping detections and introducing a minimum frame count threshold (MIN_FRAMES) to reduce false positives. The system demonstrated a strong balance, effectively counting objects in challenging conditions and achieving significant performance improvements over the baseline and a state-of-the-art method. For the experiments, our system led to a 91.43% average improvement in RMSE compared to the Ultralytics object counting method and 60.53% improvement compared to our baseline without NMS and MIN_FRAMES. Future work should focus on improving multi-camera setups, enhancing spatial alignment, and refining object identity tracking across different viewpoints. Incorporating techniques like attention mechanisms and optimizing the system for real-time execution on edge devices could broaden its applications in traffic analysis or crowd management.

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